Lab 7 – NumPy

1. Overview

In this lab assignment, you will explore the fundamentals of NumPy using a small stock dataset covering 20 business days. The dataset includes the following columns: Date, Open, Close, and Volume.

You will use NumPy to load, manipulate, and analyze the stock data. Through array creation, indexing, vectorized arithmetic, Boolean masking, and statistical calculations, you will analyze daily stock performance, identify the best and worst trading days, and calculate an estimated cumulative return.

Use NumPy as your primary tool and write clean, well-commented Python code.

2. Learning Objectives

By completing this lab, you will be able to:

- Load CSV data using np.genfromtxt().
- Extract and manipulate columns from structured NumPy arrays.
- Perform vectorized arithmetic operations.
- Calculate statistical measures using NumPy.
- Apply Boolean masking to analyze data.
- Identify maximum and minimum values using NumPy functions.
- Calculate cumulative returns using NumPy operations.

3. Tasks

Task 1 – Load and Extract the Stock Data

Use np.genfromtxt() to load the stock CSV file.

Requirements:

- Use delimiter=",".
- Use names=True.
- Use dtype=None.
- Use encoding="utf-8".

Extract the following columns into separate NumPy arrays:

- date
- openp
- closep
- vol

Display the extracted arrays or their contents to verify that the data was loaded correctly.

Task 2 – Calculate Daily Returns

Calculate the daily open-to-close return (rate) as a percentage using the following formula: rate = (Close - Open) / Open * 100

Requirements:

- Use vectorized NumPy arithmetic.
- Store the calculated values in an array named rate.
- Print the first five daily rates with appropriate labels.

Task 3 – Calculate Basic Statistics

Analyze the daily return rates using NumPy statistical functions.

Calculate:

- Mean using np.mean().
- Median using np.median().
- Standard deviation using np.std().

Display all three results with appropriate labels, rounded to two decimal places.

Task 4 – Identify the Best and Worst Trading Days

Identify the trading days with the highest and lowest daily returns.

Use:

- np.argmax(rate) to identify the best trading day.
- np.argmin(rate) to identify the worst trading day.

For each day, display:

- Date
- Opening price
- Closing price
- Daily return rate (%)

Task 5 – Analyze Up and Down Trading Days

Determine how many trading days had positive and negative returns.

Requirements:

- Use a Boolean mask with rate > 0 to identify positive-return days.
- Use a Boolean mask with rate < 0 to identify negative-return days.
- Use np.sum() to count each category.
- Display both counts with appropriate labels.

Task 6 – Calculate Cumulative Return

Calculate the estimated cumulative return using the daily open-to-close return rates.

Requirements:

1. Convert each daily rate into a return factor using: f = 1 + rate / 100
2. Calculate the cumulative return using: cum_return = (np.prod(f) - 1) * 100
3. Use np.prod() to calculate the product of the daily return factors.
4. Print the cumulative return as a percentage, rounded to two decimal places.

4. Present and Demonstrate Your Lab Solution

After completing and testing your solution, prepare to demonstrate your work to the instructor during the Thursday lab session.

Ensure that your Python source file and the stock CSV data file are available and ready for your demonstration.

A. Demonstrate Your Solution

Present your solution step by step and explain how it works.

Your demonstration must include:

- Loading the CSV file using np.genfromtxt().
- Extracting the four columns into separate NumPy arrays.
- Calculating and displaying the first five daily return rates.
- Calculating the mean, median, and standard deviation.
- Identifying the best and worst trading days.
- Using Boolean masks to count positive and negative trading days.
- Calculating the cumulative return using np.prod().

Show important intermediate results so the instructor can verify that your solution satisfies all six tasks.

B. Test Your Solution

Verify that your program produces the expected results.

Your tests must confirm that:

- The CSV file loads correctly and contains 20 business-day records.

- The four extracted arrays contain the correct data.
- Daily return rates are calculated using the correct formula.
- The first five daily rates are displayed correctly.
- The statistical calculations produce the correct results.
- The best and worst trading days are identified correctly.
- Positive and negative trading days are counted correctly.
- The cumulative return is calculated correctly.

For each test, show the relevant input or intermediate data, the expected result, the actual result, and whether the test passed.
Correct any unexpected results before your demonstration.

5. Submission Requirements

To receive marks, you must **complete, demonstrate, and submit** your laboratory work.

A. Complete and Demonstrate

- Complete all six tasks.
- During the **Thursday lab session**, demonstrate your completed solution to the instructor for assessment.
- Be prepared to explain your NumPy operations and results.

B. Prepare Your Submission

After completing your demonstration, submit **one ZIP file** to the designated Lab 7 folder in D2L containing:

- Your Python source file (.py).
- The required stock CSV data file.
- Any additional files required to run your program.

Use the following naming format:
lastname-firstname-Lab-7.zip

C. Submission Deadline

Submit your ZIP file by the deadline specified in D2L.
Important: Both the in-person demonstration during the **Thursday lab session** and the D2L submission are required to receive marks.

Note: The cumulative return calculated in Task 6 is a quick estimate based on multiplying daily open-to-close return factors. It is not the exact return from the first day's opening price to the final day's closing price.